SCIENTIST GUIDE

Ratio Milpa 0.1.0 · Ratio Complexus

# Local knowledge as a *complex adaptive system*: model, equations and evidence

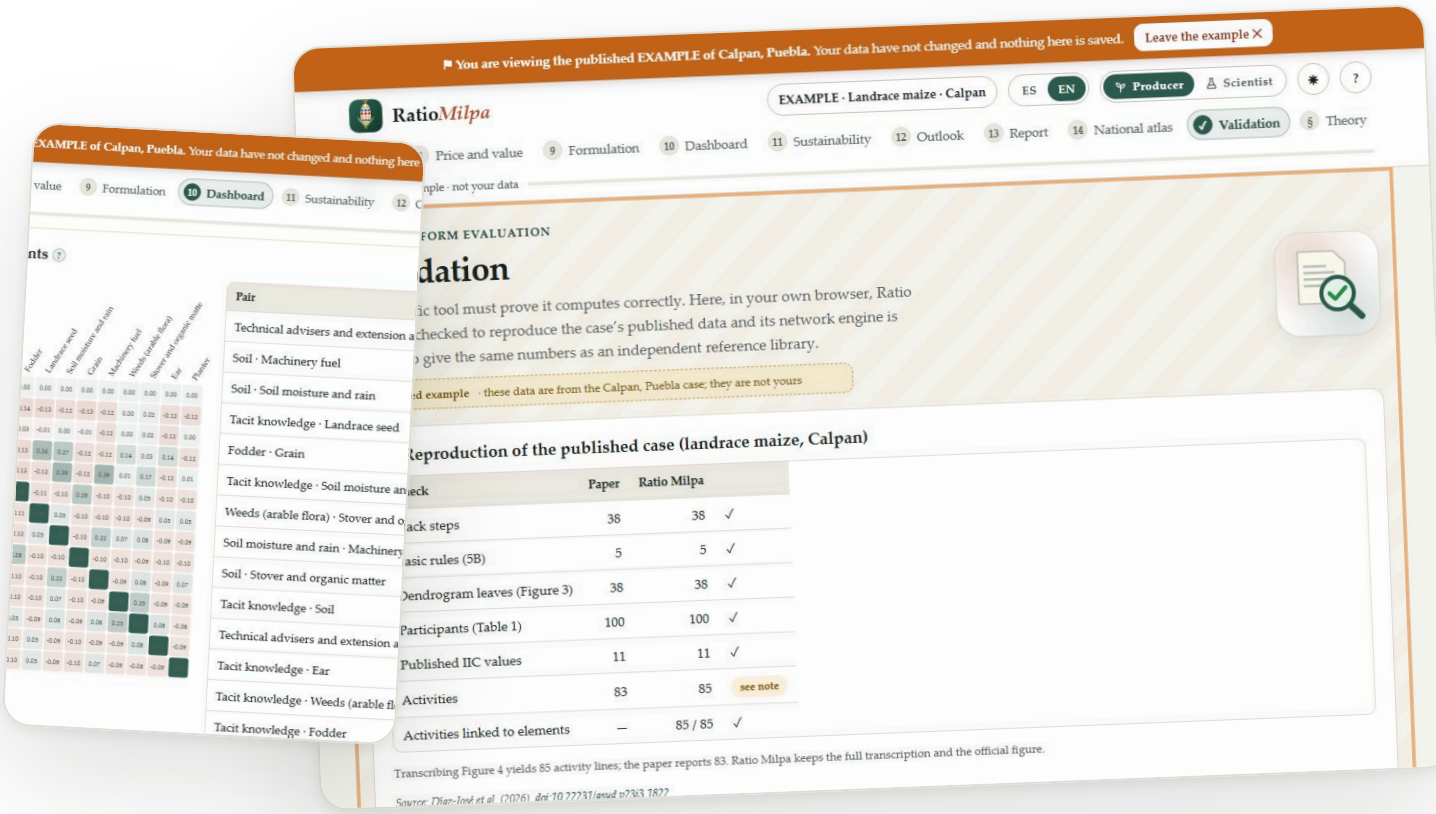
Technical documentation of Ratio Milpa for research and teaching: the data model, the weighted network and its statistics, the cladogram, the sustainability, regeneration and risk indices, the outlook, validation against NetworkX and SciPy, and how to reproduce every number.

**1st edition** · October 2026 · covers Ratio Milpa 0.1.0

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*For step-by-step use, see the Producer Guide.*




1 Maize · 2 Region · 3 Requirements · 4 Management · 5 Regeneration · 6 Risks · 7 Elements · 8 Price and value · 9 Formulation · 10 Dashboard · 11 Sustainability · 12 Outlook · 13 Report · 14 Atlas

FRONT MATTER

# Scope, data provenance and conventions

Ratio Milpa represents a crop or livestock farm as a complex adaptive system (Holland, 1995): its elements are nodes and its management activities, the links. On that network it computes centralities, communities, correlations and a cladogram; assesses sustainability, regeneration and risk; and projects scenarios at 2, 5 and 10 years. Everything runs in the browser, in a single HTML file with no external dependencies.

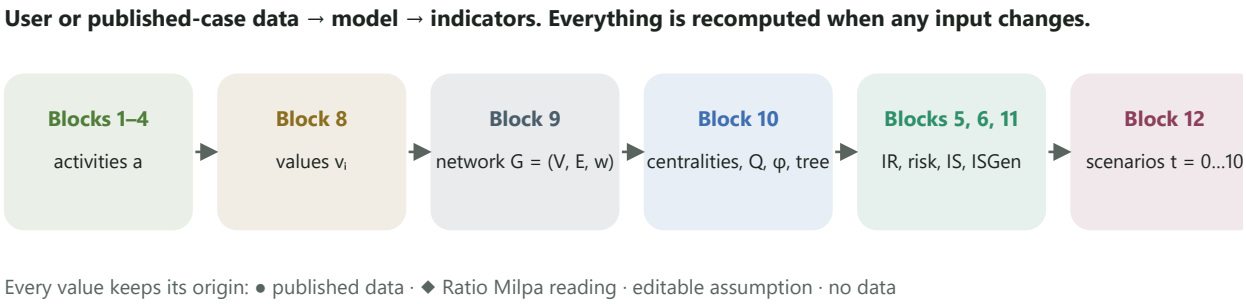



**FIGURE P.1** **The computation flow.** From activities to all other indicators.

# Provenance of every value

| LABEL | MEANING |
|---|---|
| **● Dato publicado** | Taken from a cited source: Díaz-José et al. (2026), Mojica Zárate (2017), INEGI (2020), CONABIO (2020), IPCC (2006, 2018, 2019), Zhao et al. (2017). |
| **◆ Lectura de Ratio Milpa** | Rule, grouping or value proposed by the app (for example, activity–element links, impact profiles, class cut-offs). Editable. |
| **Supuesto** | Scenario parameter with no published basis (prices, costs, labour, risk reduction factors). Editable and flagged. |
| **Sin dato** | The value is not in the sources and was not entered by the user. The app does not impute it. |

| | | | |
|---|---|---|---|
| REFERENCE CASE | Landrace maize, San Andrés Calpan, Puebla (21026). 38 steps, 83 activities, 5 rules; 100 informants, 2022–2024. | ELEMENTS | 63 in 10 categories, from the paper; those inferred from an activity are flagged. |
| TERRITORY | 32 states, 2,478 municipalities, 54,357 localities (INEGI, 2020). | IMPLEMENTATION | Library-free JavaScript; one file built from `src/` with `armar.py`. |

METHOD

# 1 Data model

Activities, elements, value and impact: the inputs to every computation.

## 1.1 Activities and incidence

Each activity $a$ belongs to a step $p$ and, when there is a Stack, to a 5B rule. It has a type (traditional or innovated) and an element set $E_a$. The incidence matrix $X$ (elements × activities) is 1 if the element takes part. Stack activities are questions: only those the user answers "yes" enter the computation; those answered "no" and unanswered ones stay out (nothing is preloaded). In the Calpan case the app activates 85 activities.

## 1.2 Element value

The user rates each element on four dimensions $k \in$ {economic, use, cultural, ecological}, from 0 to 5, with weights $\omega_k \in [0, 3]$. Without a rating the structural value, relative participation, is used:

$$v_i = \Sigma_k\, \omega_k \cdot c_{ik} / \Sigma_k\, 5\, \omega_k \quad \text{o bien} \quad v_i = n_i / \max_j n_j \tag{1}$$

where $c_{ik}$ is the rating and $n_i$ the number of activities element $i$ takes part in.

## 1.3 Impact by compartment

Five compartments $c$: soil, water, atmosphere, vegetation and fauna. Each element has an impact profile (a Ratio Milpa reading with its rationale); the activity's impact is the clipped sum, or the value edited by the user:

$$s_{ac} = \min(2, \max(-2, \Sigma_{i \in E_a}\, \pi_{ic})) \tag{2}$$

METHOD

# 2 The weighted network and its statistics

## 2.1 Network construction

Two elements are linked when they co-occur in an activity; the weight accumulates the geometric mean of their values over all shared activities, plus the structural ties documented in the paper (for example, local academia with the actors):

$$w_{ij} = \Sigma_{a\,:\,i,\,j \in E_a} \sqrt{(v_i \cdot v_j)} \qquad (3)$$

A zero value is replaced by 0.05 so the link is not cancelled. The “hide the omnipresent node” option excludes farmers (PM), who take part in almost every activity.

## 2.2 Centralities

| MEASURE | DEFINITION IN THE APP |
|---|---|
| Weighted degree | $s_i = \Sigma_j\, w_{ij}$ |
| Betweenness | Brandes (2001) with Dijkstra and length $1/w_{ij}$; unnormalised (Freeman, 1978). |
| Closeness | Inverse of mean 1/w distance to reachable nodes, scaled by the reachable fraction (Wasserman & Faust, 1994). |
| Eigenvector | Power iteration on the weighted matrix (200 iterations, max normalisation). |
| Clustering | Unweighted local coefficient; mean path and diameter by BFS (Watts & Strogatz, 1998). |

## 2.3 Communities and modularity

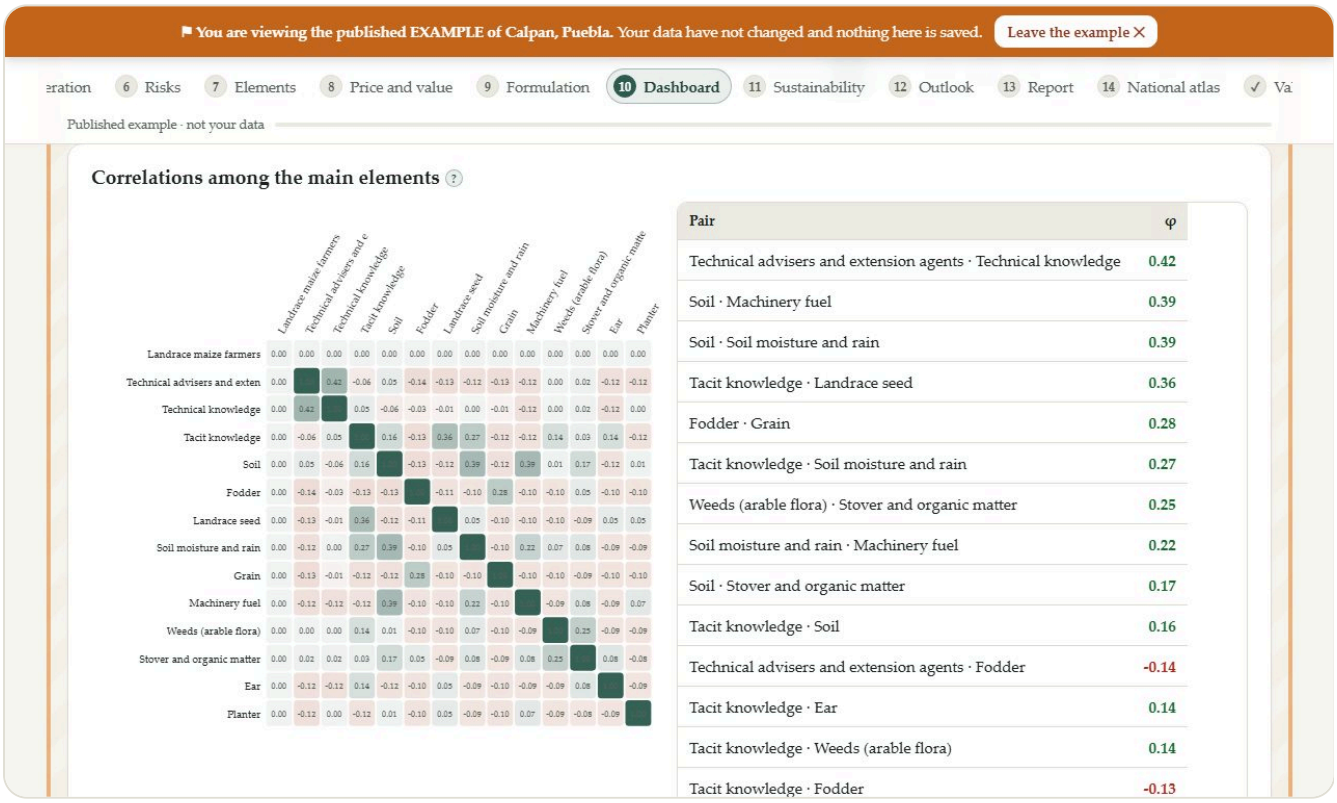
Communities are detected with the Louvain method (Blondel et al., 2008), deterministic: local moves maximising ΔQ and aggregation, repeated until Q stops improving. Modularity (Newman, 2006):

$$Q = (1/2m)\ \Sigma_{ij}\ [\ w_{ij} - s_i\, s_j/2m\ ]\ \delta(c_i,\ c_j) \tag{4}$$

## 2.4 φ correlation

Pearson correlation between two elements' incidence vectors across the $N$ activities (equivalent to the φ coefficient of a 2 × 2 table):

$$\varphi_{ij} = \Sigma_a (x_{ia} - \bar{x}_i)(x_{ja} - \bar{x}_j)/(N\ \sigma_i\ \sigma_j) \tag{5}$$




| Pair | φ |
|---|---|
| Technical advisers and extension agents · Technical knowledge | 0.42 |
| Soil · Machinery fuel | 0.39 |
| Soil · Soil moisture and rain | 0.39 |
| Tacit knowledge · Landrace seed | 0.36 |
| Fodder · Grain | 0.28 |
| Tacit knowledge · Soil moisture and rain | 0.27 |
| Weeds (arable flora) · Stover and organic matter | 0.25 |
| Soil moisture and rain · Machinery fuel | 0.22 |
| Soil · Stover and organic matter | 0.17 |
| Tacit knowledge · Soil | 0.16 |
| Technical advisers and extension agents · Fodder | -0.14 |
| Tacit knowledge · Ear | 0.14 |
| Tacit knowledge · Weeds (arable flora) | 0.14 |
| Tacit knowledge · Fodder | -0.13 |

**FIGURE 2.1** **φ matrix** of the 14 highest-degree elements (scientist view).

**CALPAN**

Full network: 61 nodes, 209 links, density 0.114, Q = 0.038 (4 communities). Without PM: 60 nodes, 158 links, Q = 0.4496 with 7 communities, mean clustering 0.5252 and mean path 2.9017. The highest betweenness without PM belongs to technical advisers, technical knowledge, tacit knowledge and soil.

METHOD

# 3 Nearest-neighbour cladogram

## 3.1 Network distance

The distance between two elements is the shortest-path length in the full network, with length $1/w_{ij}$ per link (the same metric as betweenness and closeness). The path may pass through any element even if the tree shows a subset (actors; actors and institutions; all). Pairs with no path get 1.15 times the largest observed distance.

$$d_{ij} = \min_{\text{caminos } i \to j} \Sigma_{(u,v)} 1/w_{uv} \tag{6}$$

## 3.2 Single linkage and cophenetic correlation

Agglomerative hierarchical clustering by single linkage (Sneath, 1957; Sokal & Sneath, 1963): at each step the two groups with the closest pair of members merge, $D(A, B) = \min_{i \in A,\, j \in B} d_{ij}$. Tree fidelity is measured with Sokal and Rohlf's (1962) cophenetic correlation, the Pearson correlation between $d_{ij}$ and the height at which $i$ and $j$ merge.

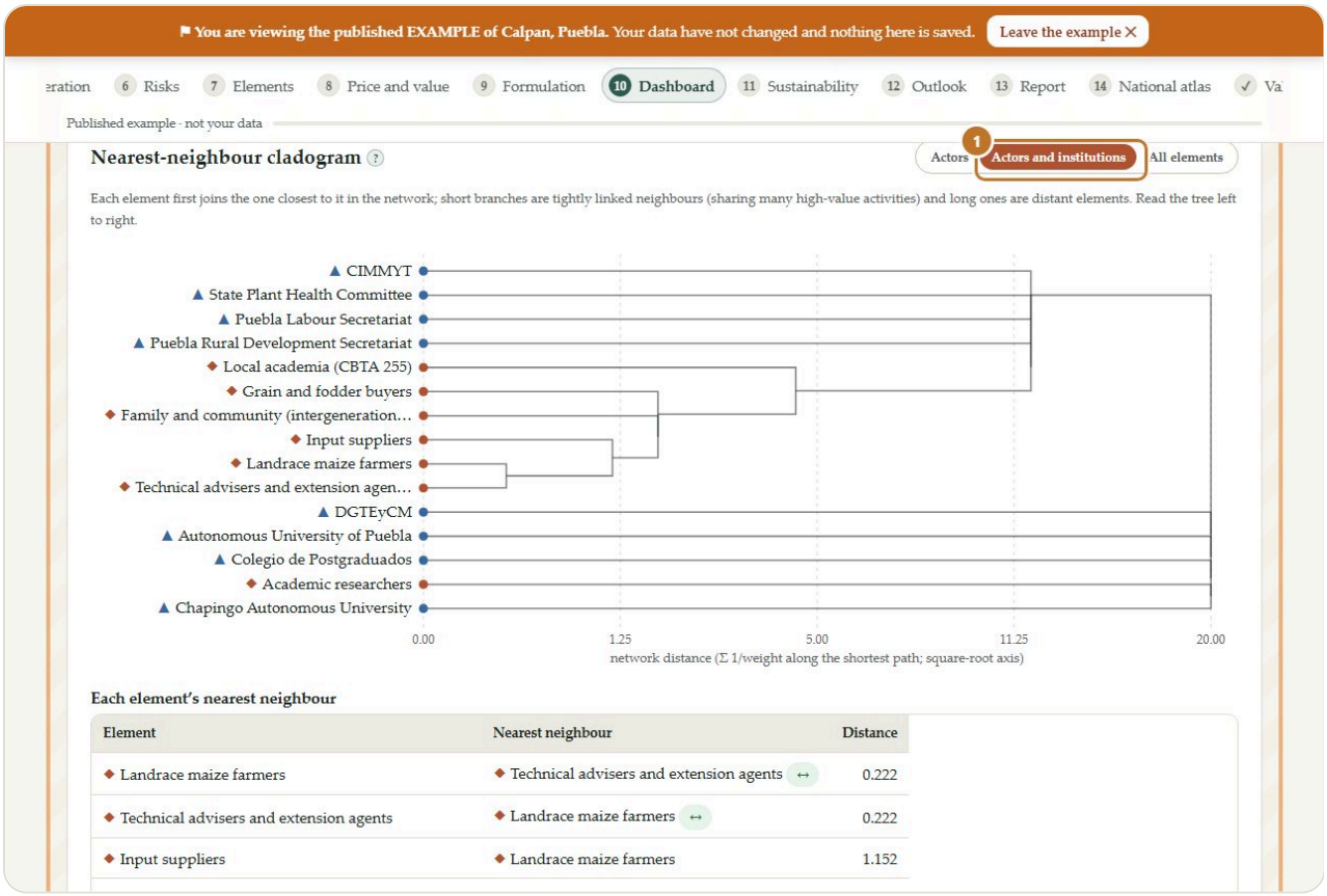



**FIGURE 3.1** **Calpan actors and institutions cladogram.** Square-root axis; cophenetic correlation 0.853.

| SUBSET | N | COPHENETIC R | \|Δ\| VS SCIPY |
|---|---|---|---|
| Actors | 7 | 0.9917 | $1.1 \times 10^{-16}$ |
| Actors and institutions | 15 | 0.8526 | $1.0 \times 10^{-15}$ |
| All elements | 61 | 0.9103 | $2.1 \times 10^{-15}$ |

**?** **WHY SINGLE LINKAGE?**

Because it directly answers "who is whose nearest neighbour?": each element's first merge is with its nearest neighbour, and the tree matches the minimum spanning tree of the distances.

METHOD

# 4 Sustainability, SDGs and carbon footprint

## 4.1 Activity sustainability index

$$IS_a = (1/5)\ \Sigma_c\ (s_{ac} + 2)/4 \quad \in [0, 1] \tag{7}$$

0.5 is neutral. The system IS is the mean of its activities; it is also reported by step and by type (traditional or innovated). In Calpan: 0.51.

## 4.2 Sarandón's general index

Following Sarandón (2002), each dimension is expressed on 0–4 through sub-indicators; their translation into app data is a Ratio Milpa reading:

| DIMENSION | SUB-INDICATORS (0–4) | CALPAN |
|---|---|---|
| IK | A margin 4(I − C)/I · B distinct products (cap 4) · C 4(1 − steps with external inputs/steps) | 3.11 |
| IE | Mean of ($s_{ac}$ + 2) across activities and compartments | 2.05 |
| ISC | A rules with local knowledge · B rules with technical knowledge · C linked institutions | 2.67 |

$$ISGen = (IK + IE + ISC)/3\ ; \quad \text{sostenible si } ISGen > 2 \text{ y } \min(IK, IE, ISC) \geq 2 \tag{8}$$

In Calpan, ISGen = 2.61: sustainable. The margin (IK-A) stays as no data until income and costs are entered; IK averages the available sub-indicators.

## 4.3 SDG attainment

For each SDG the available evidence is averaged: environmental (mean of $(s_{ac} + 2)/4$ over the linked compartments: SDGs 2, 6, 12, 13 and 15) and structural (activities with technical knowledge or advice, SDG 4; linked institutions, SDG 17; margin, SDGs 1 and 8). Without evidence, "no data". The compartment–SDG mapping is a Ratio Milpa reading. In Calpan 7 of the 17 have evidence.

## 4.4 Carbon footprint per hectare

$$\mathrm{CO_2e} = 2.68\,D + N\,(0.010 + 0.11 \times 0.010 + 0.24 \times 0.011) \times {}^{44}\!/_{28} \times 273 + 0.20\,U \times {}^{44}\!/_{12} \tag{9}$$

$D$: litres of diesel; $N$: kg of N applied; $U$: kg of urea. IPCC default factors (2006; 2019 refinement): EF1, FracGASF × EF4 and FracLEACH × EF5; $N_2O$ GWP100 = 273 (AR6, Forster et al., 2021). Divided by yield it gives kg $CO_2e$ per tonne.

METHOD

# 5 Regeneration index

Eight ordinal indicators $s_k \in \{0, \ldots, 4\}$ group the four USDA-NRCS soil health principles (minimal disturbance, soil cover, diversity, living roots) and the processes the regenerative agriculture literature adds (livestock integration, fewer synthetic inputs, organic matter; Newton et al., 2020; Schreefel et al., 2020), plus the Stack's local knowledge.

$$\mathrm{IR} = \Sigma_{k=1}^{8} s_k / 32 \quad \in [0, 1] \tag{10}$$

## 5.1 Suggested-value rules

| INDICATOR | RULE (RATIO MILPA READING) | CALPAN |
|---|---|---|
| Minimal tillage | n activities with plough, subsoiler or harrow (decisions excluded): 0→4, 1→3, 2→2, 3–4→1, ≥5→0 | 1 (n = 4) |
| Soil cover | 2 if there are stubble activities with positive soil impact; otherwise 1 | 2 |
| Diversity | species by system (milpa 3, MIAF 4…): 1→0, 2→1, 3→2, 4–5→3, ≥6→4 | 2 |
| Living roots | 3 if the system has perennials; otherwise 2 | 2 |
| Livestock integration | 3 with animals; 2 if fodder is produced; otherwise 0 | 2 |
| Fewer synthetics | round(4 (1 − S/(S + B))) | 0 (S = 16, B = 1) |
| Organic matter | 1, 2 or 3 by residue-returning activities (0, 1, ≥2), +1 with soil testing | 4 |
| Local knowledge | min(4, round(8 f)), f = share of activities with tacit or cultural knowledge, family or own seed | 2 (f = 0.19) |

With the suggested values, IR = 15/32 = 0.469 (47, conventional). Classes: < 0.25 degrading; 0.25–0.49 conventional; 0.50–0.74 regenerative transition; ≥ 0.75 regenerative. Trajectory is measured as ΔIR from a saved baseline and, with soil tests, as Δ% organic matter.

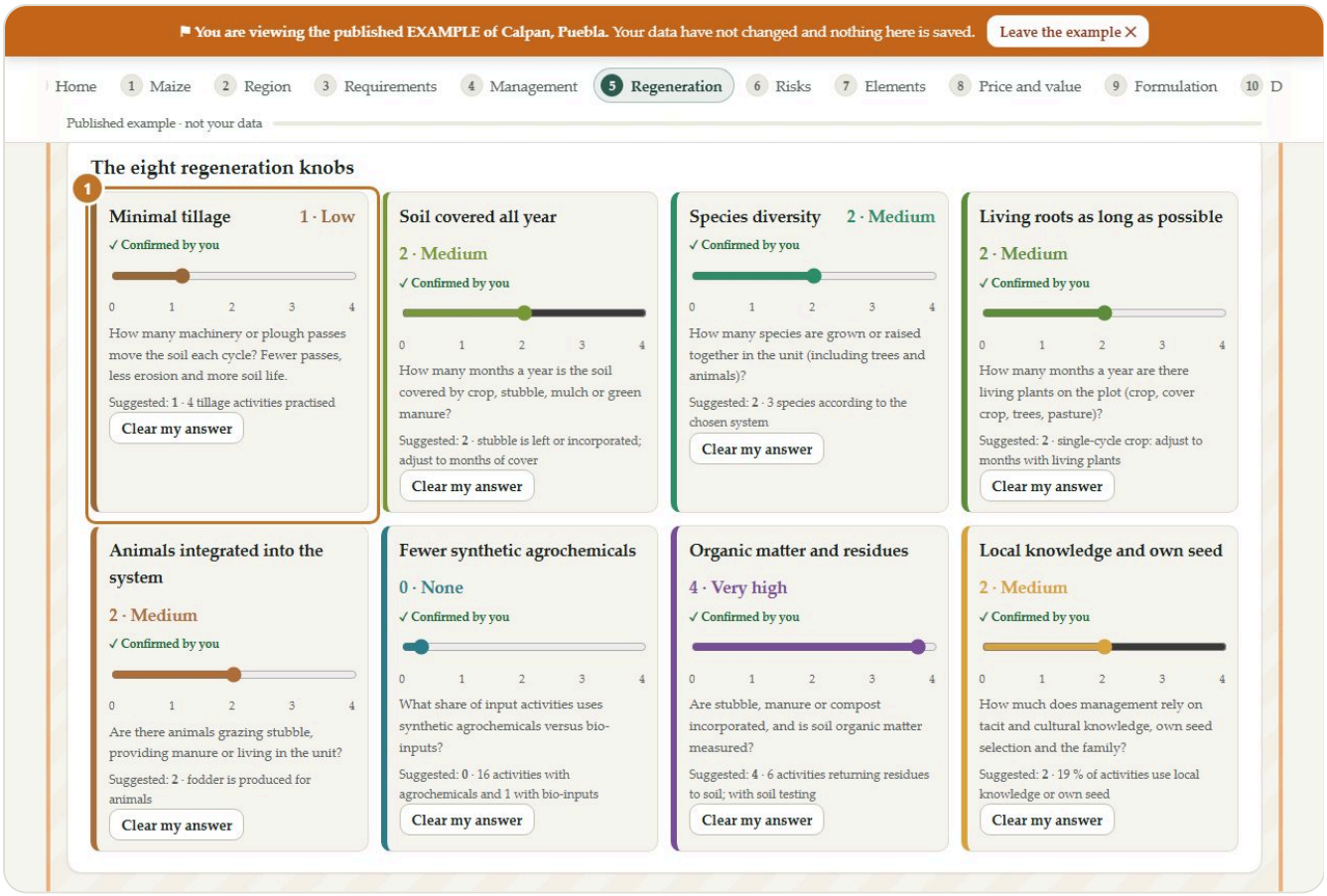



**FIGURE 5.1** **The eight indicators** with their suggested value and reason.

METHOD

# 6 Risk and adaptive capacity

Nineteen hazards $h$ in six categories. Each has likelihood $P_h$ and impact $I_h$ on 1–5 (5 × 5 matrix; IEC 31010:2019, within the ISO 31000:2018 process) and a control $c_h \in \{0, 1, 2, 3\}$. Following the IPCC (2014) framework, risk depends on the hazard and on vulnerability, which adaptive capacity reduces:

$$r_h = P_h\, I_h\, (1 - 0.4\ \mathrm{CA})(1 - 0.15\ c_h)\ ; \quad \text{índice} = \Sigma_h\, r_h / 25H \tag{11}$$

$$\mathrm{CA} = \mathrm{mean}(\mathrm{IR},\ \mathrm{IK_B}/4,\ \mathrm{ISC_B}/4,\ \mathrm{ISC_C}/4) \tag{12}$$

The 0.4 and 0.15 factors are Ratio Milpa assumptions. Levels of $r$: ≤ 4 low, 5–9 medium, 10–15 high, ≥ 16 very high. Suggested values derive from the trend scenario (warming, costs, price, labour) and the share of activities with purchased inputs or local knowledge; the rest stays at 3.

## 6.1 Hazard catalogue

Each hazard is linked to the Stack elements that transmit it; the link is a Ratio Milpa reading and is used to show risk in the Elements and Dashboard blocks.

| CATEGORY | HAZARDS | LINKED ELEMENTS |
|---|---|---|
| Climatic (6) | Drought or late rains · frost · hail · heavy rain or flooding · heatwave · cyclones and hurricanes | HUM, CLIMA, ENSO, CICL, MJO |
| Biological (3) | Pests · diseases · resistant or invasive weeds | PLAGA, ARV, HERB |
| Market (3) | Price fall · rise in input and service costs · middlemen or no buyer | PRECIO, COMP, GRANO, COSTOS, FERTG, PI, TRANS |
| Social (3) | Labour shortage and no succession · loss of local knowledge · insecurity or theft | PM, FAM, KTAC, KCUL |
| Institutional (2) | Withdrawal of programmes and support · tenure conflicts | AT, SDR |
| Natural resources (2) | Erosion and fertility loss · water scarcity | SUELO, HUM |

## 6.2 Suggested-value rules

| HAZARD | RULE |
|---|---|
| Drought, heatwave | $P = 4$ if trend warming is ≥ 0.2 °C per decade; otherwise 3 |
| Input cost rise | $P = 4$ if trend cost grows ≥ 2 %/yr; $I = 1 + \text{round}(4 \min(1, 2f_s))$, $f_s$ = share of activities with purchased inputs |
| Labour | $P = 4$ if $g_L \le -1.5$ %; 3 if $g_L < 0$; otherwise 2 |
| Price fall | $P = 4$ if the trend price falls; otherwise 3 |
| Knowledge loss | $I = 1 + \text{round}(4 \min(1, 2f_k))$, $f_k$ = share of activities with tacit, cultural or family knowledge |
| All others | $P = 3$, $I = 3$ (moderate; adjust to the locality) |

The user can switch off a hazard that does not apply (it leaves $H$) or overwrite $P$, $I$ and control; the app flags which values were adjusted.

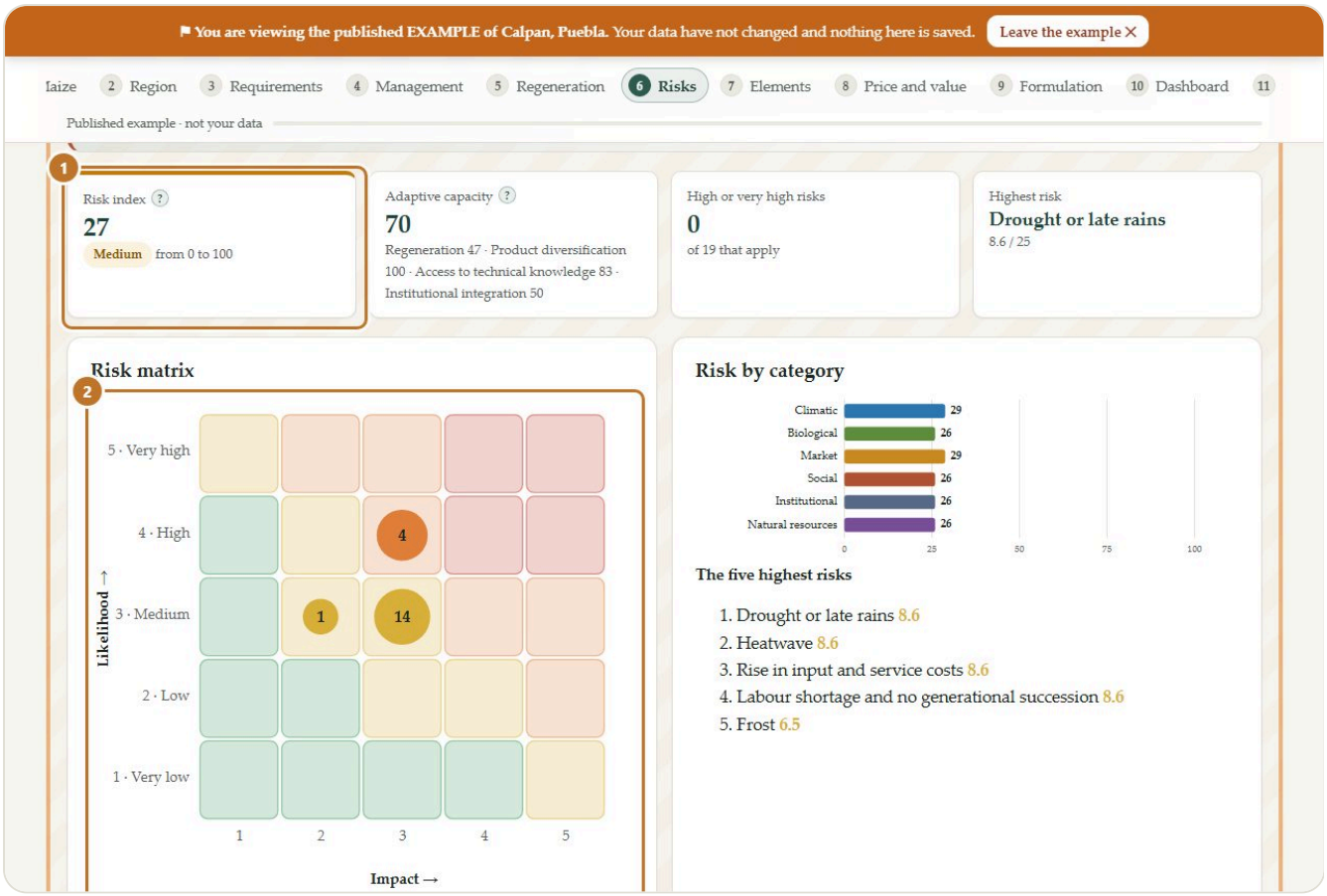



**FIGURE 6.1** **Calpan risk index and matrix.** Index 27, CA = 0.701.

◆ **CALPAN**

CA = (0.47 + 1.00 + 0.83 + 0.50)/4 = 0.70. The four highest hazards (drought, heatwave, input cost rise, labour shortage) have $P = 4$, $I = 3$ and $r = 12 \times 0.72 = 8.6$.

METHOD

# 7 Scenario model

Three exploratory scenarios (Godet & Durance, 2011) are simulated year by year, $t = 0…10$. Adoption of innovated activities follows a logistic curve (Rogers, 2003) starting from the current share $a_0$:

$$A(t) = a_0 + (A_{máx} - a_0)\,[\sigma(t - t_{50}) - \sigma(-t_{50})] / [1 - \sigma(-t_{50})] \tag{13}$$

$$Y(t) = 100\,[1 + \beta\,\Delta T\,t/10] \cdot (1 - p(t)\,L_{ef})/(1 - p_0\,L_{ef}) \cdot [1 + e_Y(A - a_0)] \cdot [1 - 0.5\,\text{déficit}(t)] \tag{14}$$

$$L_{ef} = L_{ext}(1 - 0.5\,IR_0)\;;\quad IR(t) = IR_0 + 0.5\,(1 - IR_0)(A - a_0) \tag{15}$$

β: yield change per °C (maize −7.4 %, wheat −6.0 %; Zhao et al., 2017; −4.9 % assumed for others); ΔT: warming per decade (0.1, 0.2 and 0.3 °C; IPCC, 2018 likely range); $p(t)$: extreme-year probability, linear from $p_0$ to $p_{ext}$. Price and cost grow at constant rates; available labour declines at rate $g_L$ and required labour falls with adoption. Risk at $t$ recomputes (11) with $P_{climate} + 5(p(t) - p_0)$ and CA with IR($t$). The 0.5 coefficients are assumptions.

## 7.1 Price, cost, labour and sustainability

$$Pr(t) = 100\,(1 + g_P)^t\;;\quad C(t) = 100\,(1 + g_C)^t\,(1 - s_C\,\Delta A)\;;\quad \text{rel} = 100\,(Pr \cdot Y/100)/C \tag{16}$$

$$L_d(t) = (1 + g_L)^t\;;\quad L_n(t) = 1 - r_L\,\Delta A\;;\quad \text{déficit} = \max(0,\, L_n - L_d)/L_n \tag{17}$$

$$\mathrm{IS}(t) = \mathrm{IS}_{\mathrm{trad}}\,(1 - A) + \mathrm{IS}_{\mathrm{inn}}\,A \tag{18}$$

$\Delta A = A(t) - a_0$, and it is 0 when there are no innovated activities. If the user entered income $I_0$ and cost $C_0$ per hectare (Block 8), net income is $I_0\,(Pr \cdot Y/100)/100 - C_0\,C/100$.

## 7.2 Default parameters

| PARAMETER | DESIRABLE | TREND | UNDESIRED | BASIS |
|---|---|---|---|---|
| $g_P$ real price (%/yr) | +1 | 0 | −1.5 | assumption |
| $g_C$ real cost (%/yr) | +1 | +2 | +4 | assumption |
| ΔT (°C per decade) | 0.1 | 0.2 | 0.3 | IPCC (2018) |
| $p_{\mathrm{ext}}$ extreme year by year 10 | 0.12 | 0.18 | 0.25 | assumption |
| $g_L$ labour (%/yr) | −0.5 | −1.5 | −3 | assumption |
| $A_{\max}$ maximum adoption | 0.9 | 0.5 | 0.15 | assumption |
| $t_{50}$ half-adoption year | 4 | 6 | 9 | Rogers (2003) |

Common to the three scenarios: $p_0 = 0.10$; $L_{\mathrm{ext}} = 30$ % loss in an extreme year; adoption effect on yield $e_Y = 10$ %, on cost $s_C = 5$ % and on required labour $r_L = 15$ %. All are editable assumptions in the Block 12 assumptions table.

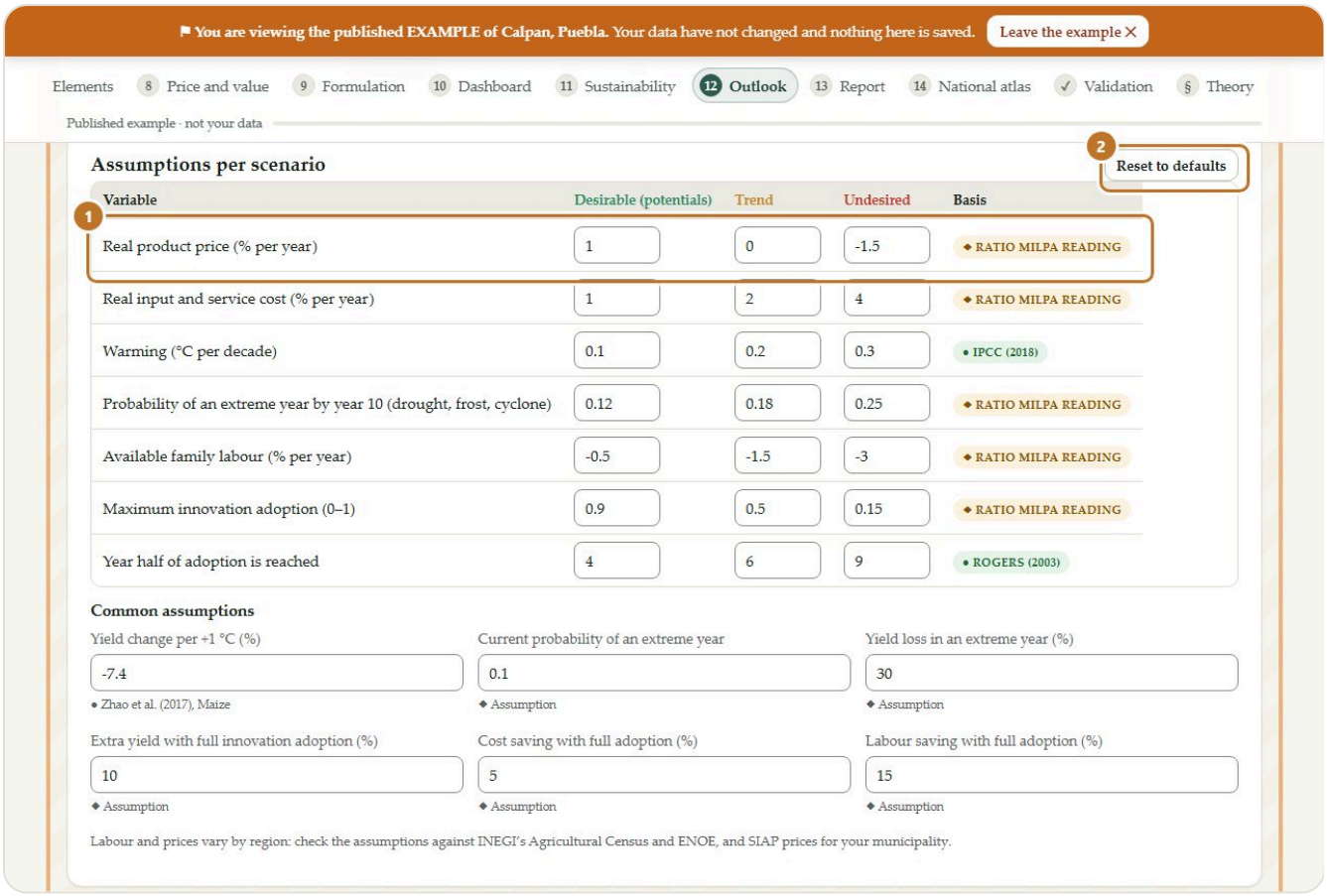



**FIGURE 7.1** **Assumptions table** with its basis: IPCC, Rogers or assumption.

| CALPAN, T = 10 (TODAY = 100) | DESIRABLE | TREND | UNDESIRED |
|---|---|---|---|
| Yield | 96 | 90 | 82 |
| Gross income ÷ cost | 96 | 74 | 48 |
| Available labour | 95 | 86 | 74 |
| Risk index (0–100) | 28 | 28 | 29 |

METHOD

# 8 Territory and map

States, municipalities and localities come from INEGI's Geostatistical Catalogue and 2020 Population and Housing Census, with population, altitude and coordinates. Regions are Banco de México's (North, North-central, Central and South). Outlines are from Natural Earth (public domain). To place each locality on the map a projection was fitted:

$$x = 32.355\,(\lambda + 102)\,\cos^{0.4375}\varphi + 514.57\,; \quad y = -34.346\,\varphi + 1122.86 \tag{19}$$

95.4 % of localities fall inside their state outline; the point is approximate.

EVIDENCE AND USE

# 9 Validation

## 9.1 Reproduction of the published case

The app checks that the loaded case matches the paper: 38 steps and 5 rules, 38 leaves in the Figure 3 dendrogram, 100 Table 1 participants and 11 published knowledge-intensity values.

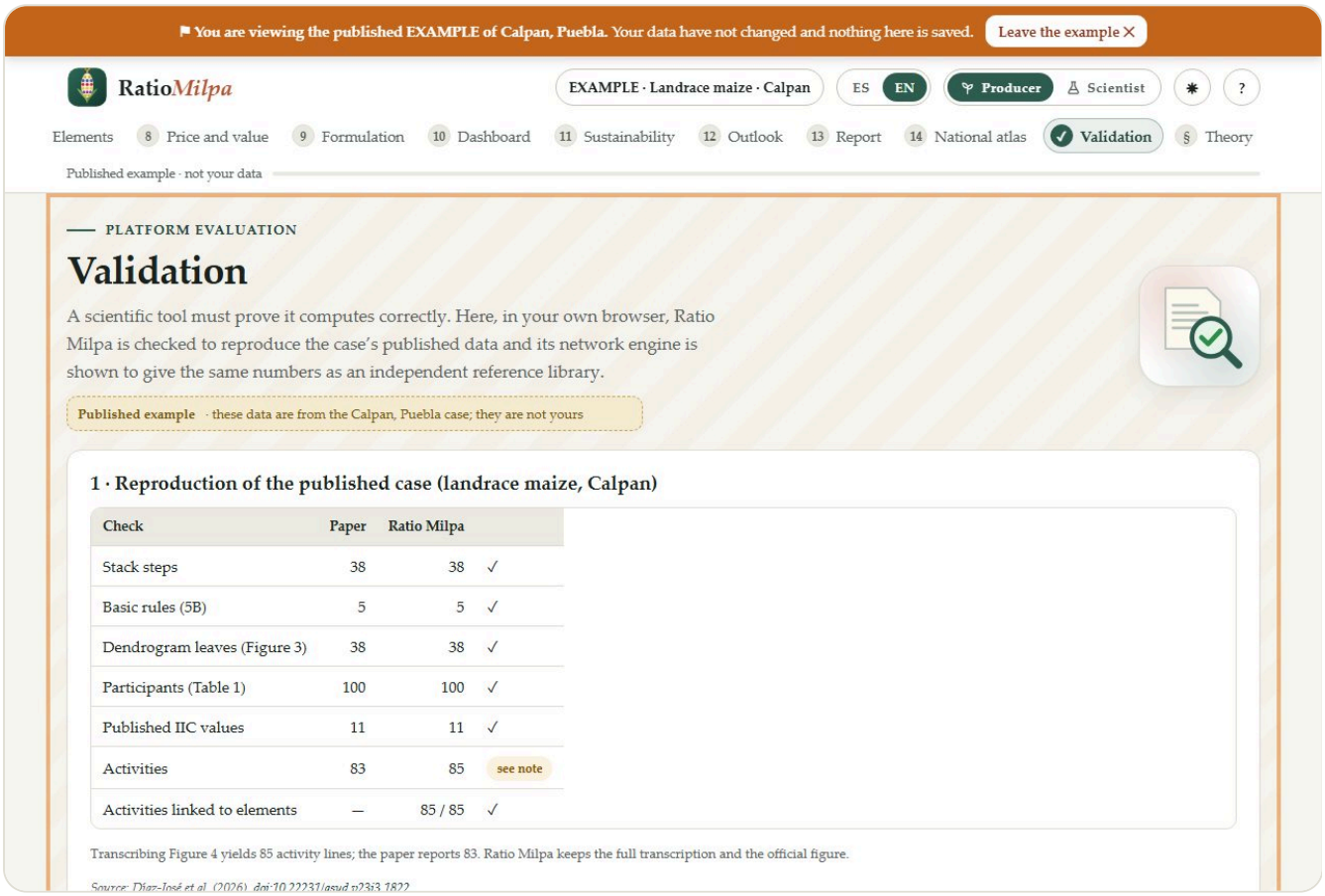



**FIGURE 9.1** **The app's validation block.**

## 9.2 Network engine against NetworkX

| TEST | MAXIMUM DIFFERENCE |
|---|---|
| Krackhardt (1990) kite: betweenness, closeness, clustering, eigenvector, mean path | < 0.00005 |
| Maize network without PM (60 nodes, 158 links): weighted betweenness | 0 |
| Partition Q (0.4496), mean clustering (0.5252), mean path (2.9017) | 0 |

For reference, NetworkX's randomised Louvain (seed 1) finds a 5-community partition with Q = 0.4422; Ratio Milpa's (7 communities, Q = 0.4496) has higher modularity.

## 9.3 Cladogram against SciPy

Merge heights and cophenetic correlation for the three subsets match `scipy.cluster.hierarchy.linkage(method='single')` and `cophenet` (SciPy 1.18) on NetworkX distances with differences below $10^{-14}$ (chapter 3 table).

EVIDENCE AND USE

# 10 Reproducibility and export

## 10.1 Code and tests

```bash
git clone https://github.com/hectortecumshe-ai/RatioMilpa
cd RatioMilpa
python armar.py                    # arma index.html desde src/
python pruebas/verificar_red.py    # red frente a NetworkX
python pruebas/verificar_clado.py  # cladograma frente a SciPy
```

## 10.2 Data for analysis

Block 13 exports to CSV the activities (with their five impacts and IS), nodes (value, participation, neighbours, weighted degree, betweenness, closeness, eigenvector, clustering, community), links (weight and number of activities), calendar, SDGs and cladogram (merges and nearest neighbours). **Save project** produces a JSON with the full state, which reproduces any result.

## 10.3 Deep links

The address accepts `#block/view/language`, for example …`/RatioMilpa/#tablero/cien/en`. This guide's screenshots were taken this way, automatically.

## 10.4 Published case and user data

Since version 0.5.0 the app starts blank. The Calpan case is loaded only in an **example mode** that is never saved (`estadoCalpan()`: maize, milpa, Calpan, sowing on day 79 and all 85 activities answered "yes"); with it the full network reproduces the values in this guide (61 nodes, 209 links; without PM, Q = 0.4496). In a user project, the structural ties documented in Calpan enter only if they touch an element of the user's activities, and Figure 4 date windows are not used as the user's dates. Suggested regeneration and risk values count in the index, flagged as provisional, until the user confirms them. Project files carry `formato: 2`; older ones are converted on opening.

## 10.5 Scope and limits

- The only case with a documented Stack is Calpan landrace maize; other crops are documented with user data.

- Activity–element links, impact profiles, the SDG mapping, regeneration rules and risk factors are Ratio Milpa readings, flagged and editable.
- Scenarios are not forecasts; their value lies in comparing explicit assumptions.
- SIAP municipal crop production will be added when its open-data service is available.

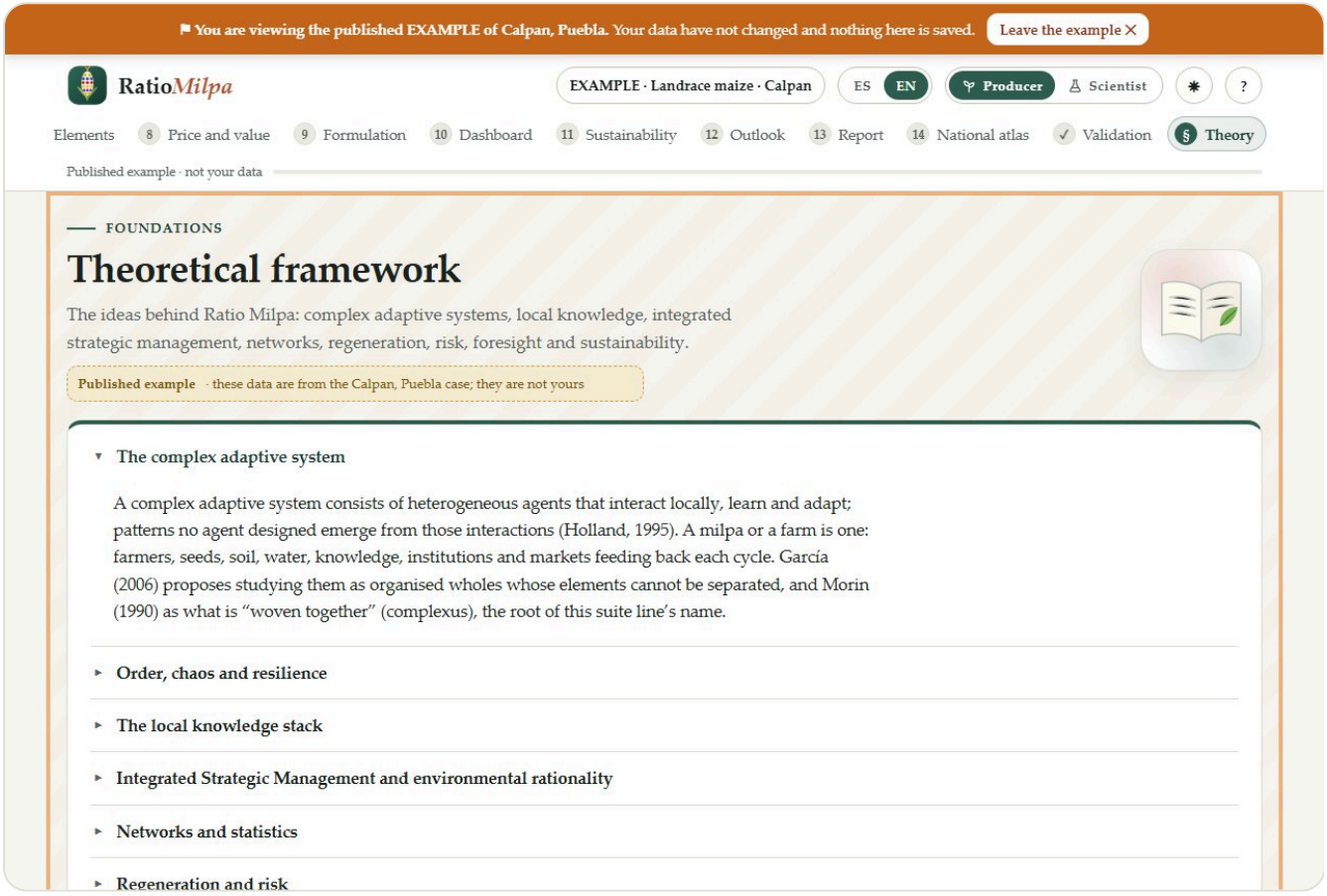



**FIGURE 10.1** **Theoretical framework** inside the app, with its bibliography.

EVIDENCE AND USE

# 11 Worked example: Calpan by hand

Every number in this chapter can be redone with a calculator and matches what app 0.4.0 shows.

## 11.1 Regeneration

$$\mathrm{IR} = (1 + 2 + 2 + 2 + 2 + 0 + 4 + 2)/32 = 15/32 = 0.469 \tag{10}$$

Class: conventional (0.25–0.49). Raising "fewer synthetics" from 0 to 2 and "tillage" from 1 to 2 would take IR to 18/32 = 0.56, in transition.

## 11.2 Adaptive capacity and drought risk

$$\mathrm{CA} = (0.469 + \tfrac{4}{4} + 3.\tfrac{33}{4} + \tfrac{2}{4})/4 = (0.469 + 1.00 + 0.833 + 0.50)/4 = 0.7005 \tag{12}$$

$$r_{\text{sequía}} = 4 \times 3 \times (1 - 0.4 \times 0.7005) \times (1 - 0.15 \times 0) = 12 \times 0.7198 = 8.64 \tag{11}$$

Medium level (5–9). With the 19 hazards active, $\Sigma r/(25 \times 19) = 0.273$: index 27. Partial control ($c = 2$) of drought would lower its $r$ to $8.64 \times 0.70 = 6.05$.

## 11.3 Sarandón general index

$$\mathrm{ISGen} = (3.11 + 2.05 + 2.67)/3 = 2.61 \;>2 \text{ y } \min = 2.05 \geq 2 \;\Rightarrow \text{sostenible} \tag{8}$$

The ecological dimension (2.05) is the closest to the threshold: it is the main lever, and it matches the low "fewer synthetics" score in IR.

## 11.4 Trend scenario at 10 years

In the Calpan base case there are no active innovated activities, so $a_0 = 0$ and $\Delta A = 0$ throughout.

| TERM | COMPUTATION | VALUE |
|---|---|---|
| Climate | $1 - 0.074 \times 0.2 \times {}^{10}\!/_{10}$ | 0.9852 |
| Effective loss | $30 \times (1 - 0.5 \times 0.469)$ | 22.97 % |
| Extreme year | $(1 - 0.18 \times 0.2297)/(1 - 0.10 \times 0.2297)$ | 0.9812 |
| Available labour | $(1 - 0.015)^{10}$ | 0.8597 |
| Shortfall factor | $1 - 0.5 \times (1 - 0.8597)$ | 0.9299 |
| **Yield** $Y$ | $100 \times 0.9852 \times 0.9812 \times 0.9299$ | **89.9** |
| Cost $C$ | $100 \times 1.02^{10}$ | 121.9 |
| **Income ÷ cost** | $100 \times (100 \times 0.899)/121.9$ | **73.7** |

The app reports 90 and 74, these values rounded. Risk at 10 years is 28: climate likelihood rises $5 \times (0.18 - 0.10) = 0.4$ points, but adaptive capacity does not change because there is no adoption.

## 11.5 Carbon footprint (illustrative example)

Hypothetical values, not from Calpan: 60 L of diesel and 100 kg of N per hectare, all as urea (217 kg; 46 % N).

$$2.68 \times 60 + 100 \times 0.01374 \times {}^{44}\!/_{28} \times 273 + 0.20 \times 217 \times {}^{44}\!/_{12} = 160.8 + 589.4 + 159.1 = 909 \text{ kg } CO_2e/\text{ha} \tag{9}$$

$0.01374 = 0.010 + 0.11 \times 0.010 + 0.24 \times 0.011$. With 3 t of grain per hectare this is 303 kg $CO_2e$ per tonne. Fertiliser $N_2O$ contributes almost two thirds.

METHOD

# 12 National atlas aggregation

## 12.1 Summary of each locality

From each `.milpa.json` file the atlas computes, with the app's own functions, a summary: location (INEGI), maize (type and race), system, yes / no answers to the 85 activities, innovated activities, IS, IR and risk (flagged provisional if the locality did not confirm all its knobs), adaptive capacity and its network links without the omnipresent node. Only the summary is stored, in the coordinator's browser.

## 12.2 Frequency of each practice

$$f_a = \Sigma_\ell\,[r_{\ell a} = \text{sí}] \,/\, \Sigma_\ell\,[r_{\ell a} \in \{\text{sí, no}\}] \tag{20}$$

$\ell$ runs over localities; unanswered items stay out of the denominator, so a missing answer is not mistaken for "no". The same ratio is computed within each Banxico region. The atlas shows next to each value how many localities answered.

## 12.3 National network

$$W_{ij} = n_{ij}/L\,; \quad V_i = n_i/L \tag{21}$$

$L$: localities with a network; $n_{ij}$: localities whose network contains link $i$–$j$; $n_i$: localities where element $i$ appears. The chapter 2 statistics are computed on this network. Regional indices are simple means of each locality's values.

## 12.4 Aggregation test

The app combines two copies of the published Calpan case. If aggregation is correct, all 85 practices are 100 % with 2 answers and the national network matches the case network (60 nodes and 158 links), with all weights equal to 1. The test runs in the Validation block and all three checks pass.

REFERENCE

# A Technical glossary

| | |
|---|---|
| **Brandes algorithm**<br>*Brandes (algoritmo)* | Exact betweenness computation in O(nm + n² log n) for weighted networks (Brandes, 2001). |
| **Adaptive capacity (CA)**<br>*Capacidad adaptativa (CA)* | Mean of regeneration and three Sarandón sub-indicators (diversification, technical knowledge, institutional integration), on 0–1. |
| **Cophenetic correlation**<br>*Correlación cofenética* | Pearson correlation between original distances and the tree's merge heights (Sokal & Rohlf, 1962). |
| **Single linkage**<br>*Enlace simple* | Agglomerative clustering in which the distance between groups is that of their closest pair of members (Sneath, 1957). |
| **Exploratory scenario**<br>*Escenario exploratorio* | Possible future built on explicit assumptions, with no assigned probability (Godet & Durance, 2011). |
| **GWP100**<br>*GWP100* | 100-year global warming potential; 273 for $N_2O$ (AR6). |
| **Incidence**<br>*Incidencia* | Binary elements × activities matrix: 1 if the element takes part in the activity. |
| **Ratio Milpa reading**<br>*Lectura de Ratio Milpa* | Value, rule or grouping proposed by the app and not published in the source; it is flagged and editable. |
| **Louvain**<br>*Louvain* | Community detection method maximising modularity through local moves and aggregation (Blondel et al., 2008). |
| **Modularity (Q)**<br>*Modularidad (Q)* | Fraction of weight within communities minus that expected at random (Newman, 2006). |
| **Foresight**<br>*Prospectiva* | Discipline building possible futures to guide present decisions (Berger; de Jouvenel; Godet). |
| **Local knowledge stack**<br>*Stack de conocimiento local* | Layered structure of farming knowledge: rules, steps and activities (Díaz-José et al., 2026). |

| | |
|---|---|
| **Structural value**<br>*Valor estructural* | An element's relative participation in activities, used as default value. |
| **Eigenvector centrality**<br>*Vector propio* | Centrality proportional to that of neighbours; obtained by power iteration. |

REFERENCE

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